# SO2R Box — Consolidated Purchase List

Two-vendor consolidation (DigiKey + Amazon). All part-number cells are clickable and resolve to the vendor search/listing for that part. Substitutions called out in *REVIEW.md* (multi-agent design review on `claude/analyze-so2r-build-plan-0xj7Q`) are baked in; the Notes column flags each swap.

## Cost summary

| Section | Subtotal |
|---|---|
| DigiKey - Control box ICs & modules | $44.46 |
| DigiKey - Connectors & hardware | $68.85 |
| DigiKey - RF box ICs, regulator, relays | $87.80 |
| DigiKey - 51 ohm 1W RF terminations | $5.40 |
| DigiKey - BPF-8A modification parts | $4.02 |
| **DigiKey subtotal** | **$210.53** |
| Amazon - Pre-built modules | $38.50 |
| Amazon - Hardware, kits, adapters | $76.00 |
| **Amazon subtotal** | **$114.50** |
| **Parts grand total (pre-tax, pre-shipping)** | **$325.03** |

**Note on totals:** the original BOM totalled ~$297. The substitutions in this list (INA333P, TC7660HCPA, USB4105-GF-A, NE8FDP, P82B96PN x2, 74HCT595N, A-taper pot) net to roughly +$5 over the BOM. The REVIEW's headline ~$580 delivered figure adds enclosures, mains-earthed PSU, PCB fab, hardware, shipping, and tax — none of those line items appear on the original BOM, so they are **not priced here**; see “Consumables & build supplies” below for the generic shopping list.

## DigiKey order

All ICs, semiconductors, parametric resistors, relays, panel connectors. Single order — DigiKey ships orders > $100 free in the US (DigiKey subtotal here is **$210.53**). Click any part number to jump to that part's DigiKey search.

### *Control box — ICs & semiconductors*

| Qty | Description | Mfr PN / Search | Unit | Ext | Notes |
|---|---|---|---|---|---|
| 3 | CD4051BE 8:1 analog mux (DIP-16) | CD4051BE | $0.57 | $1.71 | |
| 1 | CD4052BE dual 4:1 analog mux (DIP-16) | CD4052BE | $0.63 | $0.63 | |
| 1 | SN74HCT595N 8-bit shift reg (DIP-16) | SN74HCT595N | $0.66 | $0.66 | REVIEW B6: HCT (TTL) variant — 74HC595N at 5V will not latch reliably from 3.3V GPIO |
| 1 | MCP6002-I/P dual op-amp (DIP-8) | MCP6002-I/P | $0.68 | $0.68 | |
| 1 | INA333P instrumentation amp (DIP-8) | INA333P | $5.00 | $5.00 | REVIEW B4: replaces INA128PA (NRND, out-of-spec at 3.3V single supply) |
| 4 | THAT1646P14-U balanced line driver (DIP-8) | THAT1646P14-U | $5.10 | $20.40 | BOM doc-only fix: actually DIP-8 not DIP-14 |
| 1 | TC7660HCPA -5V charge pump (DIP-8) | TC7660HCPA | $2.50 | $2.50 | REVIEW B2: replaces ICL7660CPA+ (20mA limit too low for 4x THAT1646) |
| 4 | MAX3232ECPE+ RS-232 transceiver (DIP-16) | MAX3232ECPE+ | $2.13 | $8.52 | |
| 2 | P82B96PN I2C bus buffer (DIP-8) | P82B96PN | $1.26 | $2.52 | REVIEW B7 + sourcing: DIP-8 (not -D SOIC). 1x control + 1x RF box. |
| 8 | LTV-817 optocoupler (DIP-4) | LTV-817 | $0.23 | $1.84 | Consider PC817XB (CTR>=130%) per REVIEW major issue |
| | | **Subtotal** | | **$44.46** | |

### *Control box — panel connectors & hardware*

| Qty | Description | Mfr PN / Search | Unit | Ext | Notes |
|---|---|---|---|---|---|
| 1 | XLR female panel jack (mic in) | NC3FD-LX | $3.50 | $3.50 | |
| 4 | XLR male panel jack (mic out x4) | NC3MD-LX | $3.60 | $14.40 | |
| 4 | DE-9 female panel mount, solder cup (CAT RS-232) | L77SDE09SOJ4F | $1.55 | $6.20 | |
| 3 | USB-C panel receptacle, 5-pin | USB4105-GF-A | $2.50 | $7.50 | REVIEW sourcing: replaces UJ20-AF-3-SMT-TR (mid-mount SMT, not panel) |
| 11 | 3.5mm TRS panel-mount jack (HP + CW key + radio audio) | 35RAPC3BH4 | $2.50 | $27.50 | Single-SKU consolidation per BOM tip: TRS for everything; mono signals leave Ring open |
| 1 | RJ45 panel mount jack (Neutrik) | NE8FDP | $5.00 | $5.00 | REVIEW sourcing: replaces 54602-908LF (PCB-mount, not panel) |
| 8 | 2-pin PCB header 2.54mm (PTT + TX-INH) | 0022232021 | $0.25 | $2.00 | |
| 1 | 10kohm stereo audio-taper pot, 6mm shaft | PTV09A-4025F-A103 | $2.75 | $2.75 | REVIEW sourcing: -A103 = log audio taper (BOM listed -B103 = linear) |
| | | **Subtotal** | | **$68.85** | |

### *RF box — ICs, regulator, relays*

| Qty | Description | Mfr PN / Search | Unit | Ext | Notes |
|---|---|---|---|---|---|
| 2 | MCP23017-E/SP 16-bit I2C GPIO expander (DIP-28) | MCP23017-E/SP | $1.46 | $2.92 | |
| 3 | ULN2803AN 8-ch Darlington driver (DIP-18) | ULN2803AN | $0.73 | $2.19 | |
| 1 | L7805CV 5V regulator (TO-220) | L7805CV | $0.61 | $0.61 | |
| 18 | Omron G2RL-1-E-CF 12V SPDT PCB relay | G2RL-1-E-CF-DC12 | $4.56 | $82.08 | Verify -DC12 suffix (12V coil) |
| | | **Subtotal** | | **$87.80** | |

### *RF box — 51 ohm 1W terminations*

| Qty | Description | Mfr PN / Search | Unit | Ext | Notes |
|---|---|---|---|---|---|
| 18 | 51 ohm 1W metal-oxide resistor | MOX1CT52R510J | $0.30 | $5.40 | RF box NC-side termination — must be 1W. Yageo MOX1 series or similar. |
| | | **Subtotal** | | **$5.40** | |

### *BPF-8A modification (per modified unit)*

| Qty | Description | Mfr PN / Search | Unit | Ext | Notes |
|---|---|---|---|---|---|
| 9 | 2N7000 N-channel MOSFET (TO-92) | 2N7000 | $0.24 | $2.16 | |
| 9 | 1N4148 signal diode (DO-35) | 1N4148 | $0.04 | $0.36 | |
| 1 | 9-position 0.1in header (BPF mod harness, both ends) | PRT-09377 | $1.50 | $1.50 | Either end; cut from a 40-pin breakaway header |
| | | **Subtotal** | | **$4.02** | |

## Amazon order

Pre-built modules (impossible or impractical to hand-solder), connectors that ship cheaper from import sellers, and bulk passive assortment kits that beat per-line DigiKey pricing. Amazon subtotal: **$114.50**. AliExpress is typically 30-50% cheaper on these items at the cost of 2-4 week lead time — same search terms apply.

### *Pre-built modules*

| Qty | Description | Mfr PN / Search | Unit | Ext | Notes |
|---|---|---|---|---|---|
| 1 | Heltec WiFi LoRa 32 V3 (ESP32-S3, USB-C) | Heltec WiFi LoRa 32 V3 ESP32-S3 | $12.00 | $12.00 | Confirm V3 (ESP32-S3) not V2 (ESP32). REVIEW B5: buzz-test GPIO 33-38 vs PSRAM before layout. |
| 2 | CJMCU-752 SC16IS752 dual UART breakout | CJMCU-752 SC16IS752 breakout | $7.00 | $14.00 | |
| 3 | CP2102N USB-UART module (USB-C) | CP2102N USB-C module | $2.50 | $7.50 | Counterfeit risk — REVIEW suggests Adafruit/Sparkfun if reliability matters |
| 1 | WS2812B 5mm/8mm through-hole pack (>=4) | WS2812B 5mm through hole | $5.00 | $5.00 | Sold in packs of 10-20; lowest single-pack price ~$5 |
| | | **Subtotal** | | **$38.50** | |

### *Hardware, kits, adapters*

| Qty | Description | Mfr PN / Search | Unit | Ext | Notes |
|---|---|---|---|---|---|
| 1 | 16mm momentary push button, white cap (4-pack) | 16mm momentary push button white | $8.00 | $8.00 | BOM specs 4 individual @ $2 — 4-pack lands at the same total |
| 1 | 12V 1A wall adapter, 2.1mm center-positive | 12V 1A power supply 2.1mm center positive | $7.00 | $7.00 | REVIEW: prefer 3-prong/earthed unit; bond chassis to mains earth |
| 4 | SO-239 UHF panel-mount jack | SO-239 panel mount UHF connector | $2.50 | $10.00 | |
| 18 | SMA or BNC panel-mount jack (match BPF modules) | SMA panel mount bulkhead jack 50 ohm | $1.00 | $18.00 | 9 per radio x 2 radios. Pick the connector your BPF-8As use. |
| 1 | 525-pc 1/4W metal-film resistor assortment | 1/4W metal film resistor assortment kit 525 | $7.00 | $7.00 | Covers 300/470/1k/4.7k/5.1k/10k/22k/100k from BOM passives table |
| 1 | Ceramic capacitor assortment (>=100nF qty 100) | ceramic capacitor assortment kit 0.1uF | $8.00 | $8.00 | Bulk 100nF for IC bypass — BOM calls for ~40+ |
| 1 | Electrolytic capacitor kit (10/100/1000uF, 16-25V) | electrolytic capacitor assortment kit | $8.00 | $8.00 | Covers 10uF/100uF/1000uF rows in BOM |
| 1 | Small perfboard / proto board pack | perfboard prototyping board pack | $5.00 | $5.00 | Covers BPF-8A mod MOSFET board + spares for control/RF box |
| 1 | Ribbon cable (28-30cm) + JST-XH 9-pin pair | JST XH 9 pin connector ribbon cable | $5.00 | $5.00 | BPF-8A mod harness |
| | | **Subtotal** | | **$76.00** | |

## Pre-power-on advisory parts (from REVIEW.md)

These items are **not on the original BOM** and are therefore not priced or totalled. They are the additional parts the multi-agent review flagged as required before first power-on or before contest use. Source from DigiKey on the same order to avoid a second shipment.

| Issue | Add | Reason |
|---|---|---|
| B3 - INA333 DC return | 2x 100k 1/4W (already in passive kit) | IA inputs need DC bias return to REF; resistor kit covers this. |
| B6 - WS2812B level shift | 1x SN74AHCT125N (DIP-14) | Mandatory 3.3V->5V level shift ahead of NeoPixel chain. |
| B8 - I2C cable surge | 2x SP0503BAHTG TVS array | One at each end of the inter-box Cat5e for SDA2/SCL2 ESD protection. |
| B8 - Phantom-power protection | 8x 47uF 63V bipolar + 8x 5.6V zener | Block 48V phantom backwards into THAT1646 outputs (4 jacks x 2 pins). |
| Major - PTT watchdog | 1x NE555P (DIP-8) | Hardware monostable kills PTT after ~30s if firmware hangs. |
| Major - +12V protection | 1x SMAJ30A TVS + 2A polyfuse | Surge clamp + reset fuse on the barrel jack. |
| Major - BPF-8A interlock | 1x SN74HC138N (DIP-16) | 1-of-N decoder so a stuck I2C bit cannot enable two filters at once. |

## Consumables & build supplies (general suggestions, unpriced)

Not on the original BOM, so no price estimate is given. Substitute whatever you already have on the bench. The list assumes you are starting from zero.

**Enclosures.** Two project boxes (control box + RF box). Hammond 1455-series extruded aluminium is the typical choice; size to fit ~150 nets across ~50 ICs in the control box and 18 relays + 4 SO-239 in the RF box. Earthed metal chassis preferred for RF immunity and mains-earth bond.

**PCB fabrication.** REVIEW recommends a single-sided JLCPCB run instead of perfboard. KiCad netlist already exists at docs/schematics/so2r-box/so2r-box.net (after fixing the P82B96 pinout). Approx 5 boards from JLC are typical low-volume fab.

**Solder, flux, wick.** Lead-free or 60/40 rosin-core, 0.6-0.8 mm. No-clean flux pen. Solder wick for rework.

**Hookup wire.** 22-24 AWG stranded silicone, 4-6 colours. ~3 m each. 18 AWG for the +12 V power rail.

**Heat-shrink, cable lacing.** 2:1 ratio, mixed-diameter pack. Spiral wrap or lacing for harnesses.

**Cat5e patch cable.** 1-3 m solid-core for the I2C/UART link between the two boxes (RJ45 to RJ45). Keep it short enough that the 4.7 k pull-ups stay inside the 159 ns rise budget the design verified.

**Radio-side cabling.** 4x DB9 male-male RS-232 cables (CAT, length per shack), 3x USB-C cables (CAT-USB), 4x 3.5 mm TRS patch (radio audio in), 1x 3.5 mm TS (foot-PTT), 4x mic XLR patch cables (operator headsets to mic-out jacks).

**Standoffs & fasteners.** M3 brass standoff kit (5/8/10 mm), M3 + M2.5 machine screws, panel-jack lock washers, rubber feet.

**IC sockets.** DIP-8/14/16/18/28 machine-pin sockets for every through-hole IC. Cheap insurance for a hand-built rework cycle.

**Bench supplies.** Isopropyl alcohol (>=99%) for flux clean-up, anti-static mat/strap, labelmaker tape, ferrite clip-ons (one per panel 3.5 mm jack — REVIEW minor item).

**Test gear (likely already on the bench).** Multimeter, oscilloscope or USB logic analyser (for I2C bring-up), 50 ohm dummy load + power meter (mandatory before the first relay click test with an antenna connected).

### *Ordering notes*

• **One DigiKey order, one Amazon order.** Two vendors total. Tax and shipping will add ~5-10% on top of the parts subtotal.
• **Verify before paying.** DigiKey small-quantity prices drift; the figures here come from the BOM's May 2026 estimates. Click each link and re-check stock.
• **Sourcing fallbacks for counterfeits.** If the Amazon CP2102N modules look dubious on arrival, swap for Silicon Labs CP2102N-MINIEK (DigiKey, ~$15 each) — same chip, guaranteed authentic.

• **Heltec V3 GPIO conflict.** REVIEW B5 says GPIO 33-38 may be PSRAM on most current production. Buzz the Heltec out before laying out the control board; if reserved, pivot to ESP32-S3-WROOM-1U-N16R8 on DigiKey instead.